Jai S Ramani

AI & ML Engineer | Aspiring Data Analyst

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PROFILE SUMMARY

Motivated Artificial Intelligence & Machine Learning undergraduate combining strong technical foundations in predictive modeling with practical expertise in business intelligence and data analytics. Proficient in Python, SQL, exploratory data analysis, and dashboarding with Power BI, Tableau, and Excel, alongside full-stack API development with React, Flask, and Django REST. Experienced in engineering computer vision workflows, time-series price analytics, and automated reporting systems. Seeking to deliver data-driven business impact through applied machine learning and analytical decision-making.

SKILLS

Databases: SQL (SQLite, PostgreSQL), MongoDB Atlas, Supabase
Frameworks & Libraries: Pandas, NumPy, OpenCV, React, Node.js, Express, Flask, Django REST
Languages: English, Hindi, Gujarati, Kannada
Programming Languages: Python, SQL, Java, JavaScript
Soft Skills: Communication, Teamwork, Problem Solving, Adaptability
Tools & Platforms: Excel, Power BI, Tableau, Docker, Git, GitHub, Pytest, Playwright

PROJECTS

PricePilot — E-Commerce Price Intelligence & Arbitrage Platform · *Python, Flask, SQLite, Pandas, Docker, Pytest* *GitHub*

- Architected an end-to-end price intelligence & arbitrage platform tracking time-series product data across **5 Amazon regions** and benchmarking prices against **18 retail marketplaces** (Flipkart, Croma, Blinkit, Walmart).
- Engineered a **6-engine analytics suite** leveraging 30-day moving averages, price volatility modeling, and promotional bank offers to generate Deal Quality Scores, price-drop forecasts, and real-time BUY / WAIT signals.
- Built a high-performance **Flask backend** with a multithreaded daemon scheduler for autonomous polling, SQLite/CSV pipelines, and real-time alert dispatch via HTML Email, Discord Webhooks, and Telegram Bots.
- Developed an interactive dashboard utilizing **Chart.js** and **GSAP** for responsive 30-day time-series analytics, inline sparklines, split-basket cost optimization, and multi-currency normalization across **7 fiat currencies**.
- Containerized with **Docker & Docker Compose** and maintained a 100% pass rate across **59 automated unit and integration tests** with **pytest**, covering scrapers, REST endpoints, and export pipelines.

Samyukt — Cloud Version Control & Collaboration Platform · *React 19, Node.js, Express, MongoDB, WebSockets, Docker* *GitHub*

- Architected a cloud-native version control platform featuring a **3-layer semantic merge engine** (text-level diff, AST structural parsing, and AI behavioral analysis) to automate code conflict resolution.
- Engineered database schemas and indexing across **6 core entities** in MongoDB Atlas / Mongoose with connection pooling and automated exponential backoff reconnection maintaining sub-30ms ping latency.
- Integrated real-time collaborative pipelines using **WebSockets & CRDT concepts**, enabling multi-user live presence tracking, cursor synchronization, and conflict-free concurrent editing.
- Containerized frontend/backend with **Docker & Nginx reverse proxy**, implementing automated test suites via **15+ Supertest integration specs** and **10+ Playwright E2E flows** with a 100% pass rate.

PresenceIQ — AI Attendance System · *Django REST, React, MongoDB, OpenCV* *GitHub*

- Built an automated AI facial recognition attendance platform with **Django REST**, **React**, **OpenCV**, and **InsightFace**, implementing role-based access control (RBAC) across 4 user tiers.
- Architected **7 MongoDB collections** for users, enrollments, sessions, biometric face data, attendance logs, and audit trails to enable scalable document management.
- Developed modular REST APIs across 3 core services for JWT authentication, class management, and automated facial recognition attendance logging.
- Integrated **PDF/Excel reporting**, attendance analytics visualizations, and bulk roster ingestion, validating the system with 20 student test profiles across live sessions.

EDUCATION

Bachelor of Engineering in Artificial Intelligence & Machine Learning 2023 — 2027
Visvesvaraya Technological University · CGPA: 8.14 / 10 *Bangalore, Karnataka, India*
Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Machine Learning, Object-Oriented Programming

Higher Secondary Education (Class XII) 2022 — 2023
The Kingdom College · Percentage: 94.14% *Bangalore, Karnataka, India*